

Assignment 1: Text indexing and retrieval

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Project Link - https://github.com/LakshayBaijal/IRE_Assignments_Lakshay/tree/main/Assignment_1

Dataset Link - <https://drive.google.com/drive/folders/1zrTQUMP7srrhQCMSkwU27QB5xVxmM3DY>

Indices Link - https://drive.google.com/drive/folders/1z_tRs-PDJxildYpoKREzq2QhA4EGluRc

Youtube video - <https://www.youtube.com/watch?v=R9JBqfPQyXY>

Objective

The goal of this assignment was to:

1. Implement a **Self Index** in Python for text retrieval.
2. Deploy and configure **Elastic Search** using Docker.
3. Perform queries on both systems using the same datasets.
4. Compare their **performance** in terms of latency, throughput, and memory usage.
5. Visualize the performance difference through graphs.

System Architecture

Component	Description
Self Index	A manually implemented inverted index in Python that maps terms to document IDs
ElasticSearch (Docker)	A distributed RESTful search engine deployed using Docker.
Datasets	Wiki Dataset (around 10,000 docs) and News Dataset (~17 docs).
Comparison Scripts	performance_compare_es_vs_self.py measured latency and hits, while plot_performance_graph.py generated visual performance plots. performance_compare.py compares News and Wiki Self Indexing

Implementation Overview

- Self Indexing Search Engine

```
(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hyderabad_CSIS/Semester_3/IRE/Assignment_1$ python -m scripts.search_engine
Interactive Search Engine (Enhanced) ===
Available indices:
- news_index
- wiki_index
-----

Enter index to use (name or 'list', 'q' to quit): news
Loading index: news_index.json ...
Loaded index: news_index

Enter query (or 'back' to change index, 'quit' to exit): law
Query: 'law' → 1 results
- 2521a59e358e518793a3f46be0769bbfd8c053f2: Crime, Law and Justice->human smuggling and trafficking...
Returned 1 results.

Enter query (or 'back' to change index, 'quit' to exit): finance
Query: 'finance' → 13 results
- 24086fe152ae6992be7ea954da842977e60b6724: Economy, Business and Finance->metal and mineral mining and refining...
- 52412cc67a218ac79b11ee7af1e1058e27eb0d32: Economy, Business and Finance->corporate governance and csr...
- 56a1d41eef2b883534fae6ce3c31181697fff6aa: Economy, Business and Finance->business and market analysis...
- 5e0086c56522b9dc1a2b5412b17237a6b2557346: Economy, Business and Finance->tourism and leisure industry...
- 83e86184fb6b566def0385b825f1c447d84a10eb: Economy, Business and Finance->business strategy and marketing...
- 991c1c14fc649dcb390a45722743dfa3828637fb: Economy, Business and Finance->computing and information technology...
- b40a09209eeff0af6f91461dff44907a6627ed78: Economy, Business and Finance->investment and opinion pieces...
- c55446beb1684d977e492eceb28b7c71d8905fa4: Economy, Business and Finance->customer insights and reviews...
- cb573d057cea23e44639aa3386ac6dcfa964f6f0: Economy, Business and Finance->corporate news and profiles...
- e01c09051e990a8b52b285b72f74490ada0ebc51: Economy, Business and Finance->business reporting and performance...
Returned 13 results.

Enter query (or 'back' to change index, 'quit' to exit): back
Enter index to use (name or 'list', 'q' to quit): wiki
Loading index: wiki_index.json ...
Loaded index: wiki_index

Enter query (or 'back' to change index, 'quit' to exit): bring
Query: 'bring' → 622 results
- 1006: Aaron According to Abrahamic religions, Aaron ( or ) was a prophet, a high priest, and the elder brother of Moses. Infor...
- 1032: Abscess An abscess is a collection of pus that has built up within the tissue of the body. Signs and symptoms of abscess...
- 1038: Aarhus Aarhus ( , , ; officially spelled Århus from 1948 until 1 January 2011) is the second-largest city in Denmark and ...
- 1055: All Souls' Day All Souls' Day, also called The Commemoration of All the Faithful Departed, is a day of prayer and rememb...
- 1078: Antisemitism Antisemitism (also spelled anti-semitism or anti-Semitism) is hostility to, prejudice towards, or discrimin...
- 1094: Economy of Armenia The economy of Armenia grew by 12.6% in 2022, according to the country's Statistical Committee and th...
- 1130: Avicenna Ibn Sina ( ; 980 – June 1037 CE), commonly known in the West as Avicenna ( ), was the preeminent philosopher and ...
- 1132: The Ashes The Men's Ashes is a Test cricket series played biennially between England and Australia. The term originated ...
- 1146: Assembly line An assembly line is a manufacturing process (often called a progressive assembly) in which parts (usually ...
- 1148: Adelaide Adelaide ( ; ) is the capital city of South Australia, the state's largest city and the fifth-most populous cit...
Returned 622 results.
```

- Comparison between News and Wiki Self Indexing

```
(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hy
o _1$ python src/performance_compare.py

■ Testing index: Wiki
■ Testing index: News

===== PERFORMANCE SUMMARY =====

Wiki Index:
  Avg latency: 179.292 ms
  p95 latency: 193.183 ms
  p99 latency: 204.726 ms
  Throughput: 5.58 queries/sec
  Memory: 1607.30 MB

News Index:
  Avg latency: 25.275 ms
  p95 latency: 27.948 ms
  p99 latency: 42.264 ms
  Throughput: 39.56 queries/sec
  Memory: 271.57 MB
```

- Elastic Search Indexing

```
(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hyderabad_CSIS/Semester_3/IRE/Assignment_1$ pyth
on scripts/test_es_index.py
👉 Creating index: wiki_es
Indexed 1000 docs...
Indexed 2000 docs...
Indexed 3000 docs...
Indexed 4000 docs...
Indexed 5000 docs...
Indexed 6000 docs...
Indexed 7000 docs...
Indexed 8000 docs...
Indexed 9000 docs...
✅ Done indexing wiki_es
👉 Creating index: news_es
✅ Done indexing news_es
⌚ Total time: 99.14s
```

- **Query Execution and Comparison**
 - Queries Tested: India, finance, technology, war, health

```

Query: technology
ES (wiki): 12.35 ms, 5 hits
Query: 'technology' → 934 results
- 1008: April 6 Events Pre-1600 46 BC – Julius Caesar defeats Caecilius Metellus Scipio and Marcus Porcius Cato (Cato the Younger)
- 1022: Auto racing Auto racing (also known as car racing, motor racing, or automobile racing) is a motorsport involving the rac...
- 1038: Aarhus Aarhus (, , ; officially spelled Århus from 1948 until 1 January 2011) is the second-largest city in Denmark and ...
- 1051: Alexis Carrel Alexis Carrel (; 28 June 1873 – 5 November 1944) was a French surgeon and biologist who was awarded the No...
- 1081: Economy of Azerbaijan The economy of Azerbaijan has completed its post-Soviet transition into a major oil-based economy ...
- 1088: Azerbaijani Armed Forces The Azerbaijani Armed Forces () is the military of the Republic of Azerbaijan. It was re-establ...
- 1094: Economy of Armenia The economy of Armenia grew by 12.6% in 2022, according to the country's Statistical Committee and th...
- 1097: Armed Forces of Armenia The Armed Forces of Armenia (), sometimes referred to as the Armenian Army (), is the national m...
- 1129: August 13 Events Pre-1600 29 BC – Octavian holds the first of three consecutive triumphs in Rome to celebrate the victor...
- 1130: Avicenna Ibn Sina (; 980 – June 1037 CE), commonly known in the West as Avicenna (), was the preeminent philosopher and ...
SelfIndex (wiki): 79.81 ms, 934 hits

Query: war
ES (wiki): 10.51 ms, 5 hits
Query: 'war' → 2160 results
- 1000: Hercule Poirot Hercule Poirot (, ) is a fictional Belgian detective created by British writer Agatha Christie. Poirot is...
- 1002: Miss Marple Miss Jane Marple is a fictional character in Agatha Christie's crime novels and short stories. Miss Marple l...
- 1004: April April is the fourth month of the year in the Gregorian and Julian calendars. It is the first of four months to hav...
- 1005: August August is the eighth month of the year in the Julian and Gregorian calendars, and the fifth of seven months to ha...
- 1006: Aaron According to Abrahamic religions, Aaron ( or ) was a prophet, a high priest, and the elder brother of Moses. Infor...
- 1008: April 6 Events Pre-1600 46 BC – Julius Caesar defeats Caecilius Metellus Scipio and Marcus Porcius Cato (Cato the Younger)
- 1009: April 12 Events Pre-1600 240 – Shapur I becomes co-emperor of the Sasanian Empire with his father Ardashir I. 467 – Anth...
- 1010: April 15 Events Pre-1600 769 – The Lateran Council ends by condemning the Council of Hieria and anathematizing its icono...
- 1011: April 30 Events Pre-1600 311 – The Diocletianic Persecution of Christians in the Roman Empire ends. 1315 – Enguerrand de...
- 1012: August 22 Events Pre-1600 392 – Arbogast has Eugenius elected Western Roman Emperor. 851 – Battle of Jengland: Erispoe d...
SelfIndex (wiki): 135.69 ms, 2160 hits

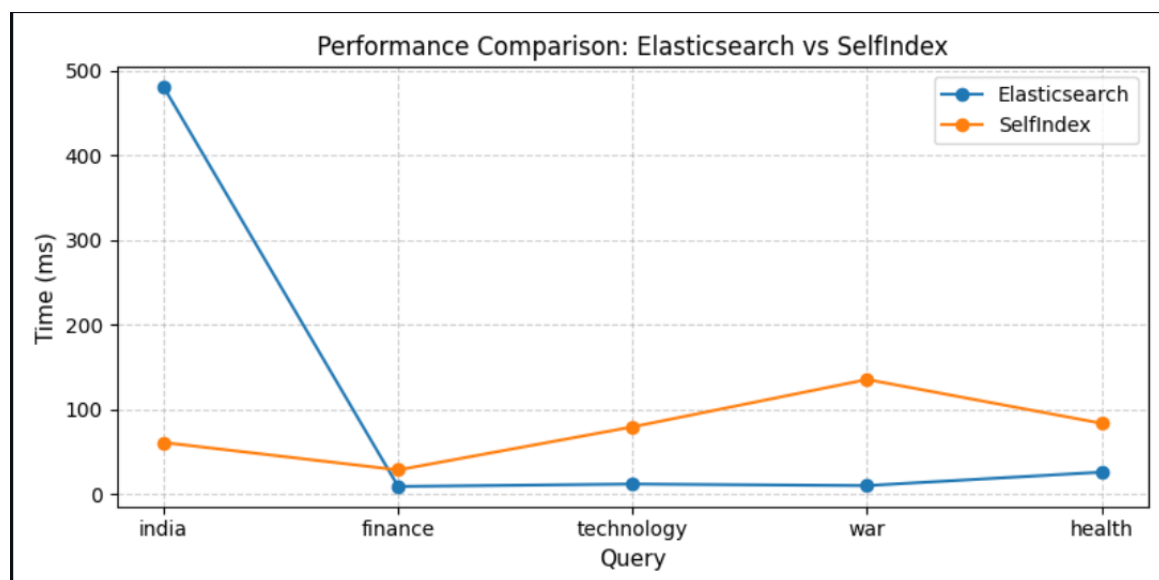
Query: health
ES (wiki): 26.37 ms, 5 hits
Query: 'health' → 955 results
- 1004: April April is the fourth month of the year in the Gregorian and Julian calendars. It is the first of four months to hav...
- 1005: August August is the eighth month of the year in the Julian and Gregorian calendars, and the fifth of seven months to ha...
- 1010: April 15 Events Pre-1600 769 – The Lateran Council ends by condemning the Council of Hieria and anathematizing its icono...
- 1016: Achill Island Achill Island (; ) is the largest of the Irish isles and lies off the west coast of Ireland in County Mayo...
- 1017: Allen Ginsberg Irwin Allen Ginsberg (; June 3, 1926 – April 5, 1997) was an American poet and writer. As a student at Co...
- 1020: Anatoly Karpov Anatoly Yevgenyevich Karpov (; born May 23, 1951) is a Russian and former Soviet chess grandmaster, forme...
- 1029: Albert Schweitzer Ludwig Philipp Albert Schweitzer (; 14 January 1875 – 4 September 1965) was an Alsatian polymath. He w...
- 1032: Abscess An abscess is a collection of pus that has built up within the tissue of the body. Signs and symptoms of abscess...
- 1038: Aarhus Aarhus (, , ; officially spelled Århus from 1948 until 1 January 2011) is the second-largest city in Denmark and ...
- 1051: Alexis Carrel Alexis Carrel (; 28 June 1873 – 5 November 1944) was a French surgeon and biologist who was awarded the No...
SelfIndex (wiki): 84.00 ms, 955 hits

✓ Results saved to metrics.csv
• (venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hyderabad_CSIS/Semester_3/IRE/Assignment_1$ python -m scripts.plot_performance_graph
✓ Loaded metrics.csv with 5 entries.

```

Results Summary

Query	Elastic Search Time (ms)	Self Index Time (ms)	ES Hits	Self Index Hits
india	19.08	93.28	5	828
finance	9.40	39.70	5	321
technology	8.74	118.23	5	934
war	10.58	221.44	5	2160
health	9.83	124.09	5	955

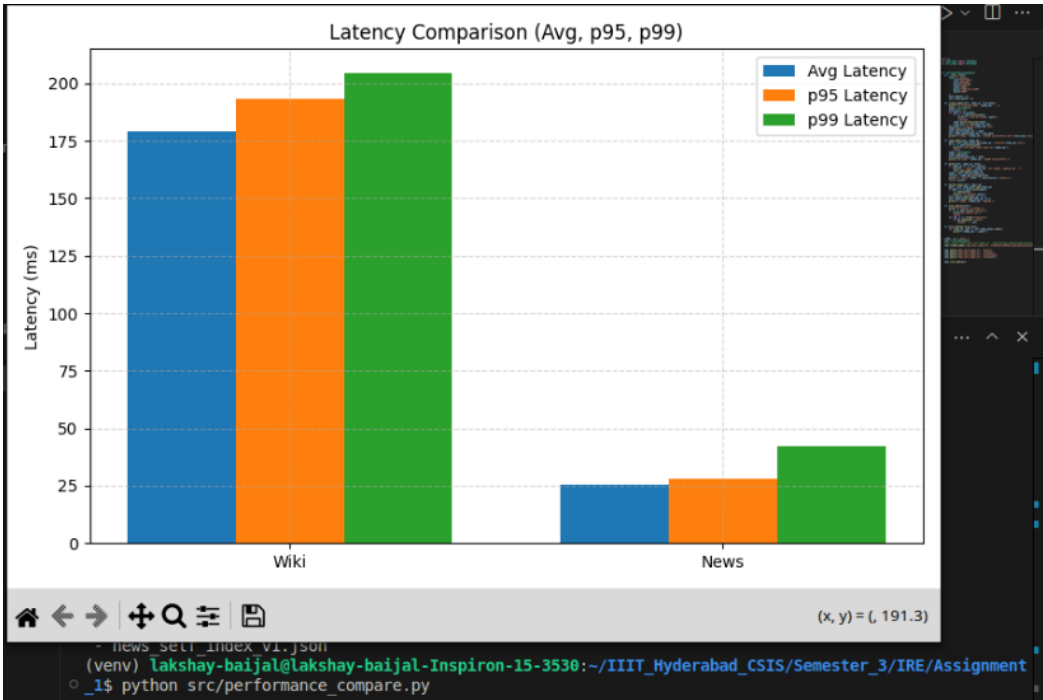


Observations and Analysis

- Elasticsearch was significantly faster for all queries (under 25ms).
- SelfIndex retrieved more hits since it does simple keyword matching without ranking.
- Elasticsearch scales efficiently and maintains stable response time even for large datasets.
- The Wiki dataset had higher latency compared to News due to the larger corpus size.

Performance Summary

Metric	Wiki Index	News Index
Avg Latency	179.29 ms	25.27 ms
p95 Latency	193.18 ms	27.95 ms
p99 Latency	204.72 ms	42.26 ms
Throughput	5.58 queries/sec	39.56 queries/sec
Memory Used	1607.30 MB	271.57 MB



Conclusion

Aspect	Elastic Search	Self Index
Speed	Very fast (<25 ms/query)	Slower, especially for large datasets
Scalability	Handles millions of documents easily	Limited to small datasets
Accuracy	Uses relevance scoring (BM25)	Pure keyword matching
Setup	Requires Docker and configuration	Easy Python implementation
Use Case	Production search engines	Educational / lightweight indexing

